

Rigid Analytic Geometry Week 3 Solution

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Problem 1

As we have seen in the previous sheet that for a K -norm $\|\cdot\|$ on L , there is $C > 0$ such that

$$\forall x, y \in L, \|xy\| \leq C\|x\|\|y\|$$

Let $\|\cdot\|'$ be another norm such that

$$\|\cdot\|' = C\|\cdot\|.$$

Then using the equation,

$$\frac{1}{C}\|xy\|' = \|xy\| \leq C\|x\|\|y\| = \frac{C}{C^2}\|x\|'\|y\|'.$$

Thus,

$$\|xy\|' \leq \|x\|'\|y\|'.$$

Problem 2

Suppose the limit,

$$|l|_L = \lim_{n \rightarrow \infty} \sqrt[n]{\|l^n\|_1}$$

exists for some norm $\|\cdot\|_1$. Then for any norm $\|\cdot\|_2$, we have,

$$c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1,$$

for some $c, C > 0$. Thus using the sandwich theorem, we obtain

$$\lim_{n \rightarrow \infty} \sqrt[n]{\|l^n\|_2}$$

also exists and equals to $\lim_{n \rightarrow \infty} \sqrt[n]{\|l^n\|_1}$. Let $a_n = \log \|l^n\|$. Then we have,

$$a_{m+n} \leq a_m + a_n.$$

Using Fekete's lemma, there is a limit $\lim_{n \rightarrow \infty} \frac{a_n}{n}$ and thus there is a limit for $\lim_{n \rightarrow \infty} \sqrt[n]{\|l^n\|}$.

Problem 3

$$|xy|_L = \lim_{n \rightarrow \infty} \sqrt[n]{\|(xy)^n\|} \leq \lim_{n \rightarrow \infty} \sqrt[n]{\|x^n\|\|y^n\|} = |x|_L |y|_L.$$

Suppose $\|x\| \leq \|y\|$ and $\|y\| \neq 0$. Then,

$$|x|_L = \lim_{n \rightarrow \infty} \sqrt[n]{\|x^n\|} \leq \lim_{n \rightarrow \infty} \sqrt[n]{\|y^n\|} = |y|_L.$$

$$|x + y|_L = \lim_{n \rightarrow \infty} \sqrt[n]{\left\| y^n \left(\frac{x}{y} + 1 \right)^n \right\|} \leq \lim_{n \rightarrow \infty} \sqrt[n]{\|y^n\|} = |y|_L,$$

Thus the strong triangle equality holds. For any $\lambda \in K$,

$$|\lambda x| = \lim_{n \rightarrow \infty} \sqrt[n]{\|\lambda^n x^n\|} = |\lambda|_K \lim_{n \rightarrow \infty} \sqrt[n]{\|x^n\|} = |\lambda|_K |x|_L.$$

Finally, $|0|_L = 0$ as $\|0^n\| = 0$.

Problem 4

This amounts to show that $|1|_L = |1|_K$. Take a supremum norm $\|\cdot\|_\infty$, then $\|1\|_\infty = 1$. Since the choice of norms is independent when defining $|\cdot|_L$, we see,

$$|1|_L = \lim_{n \rightarrow \infty} \sqrt[n]{\|1\|_\infty} = 1.$$

The general case follows that $|k|_L = |k|_K |1|_L$ for all $k \in K$.

Problem 5

From Problem 1, we have,

$$\|x\| = \|l^{-1}lx\| \leq \|m_{l^{-1}}\|_{\text{op}} \|lx\|.$$

Multiply $\frac{1}{\|m_{l^{-1}}\|_{\text{op}}^2 \|l\|}$, we get,

$$\|m_{l^{-1}}\|^{-1} \frac{\|m_{l^{-1}}\|^{-1}}{\|l\|} \|x\| \leq \frac{\|m_{l^{-1}}\|^{-1}}{\|l\|} \|lx\|.$$

Define $\|\cdot\|' = \frac{\|m_{l^{-1}}\|^{-1}}{\|l\|} \|\cdot\|$, then the above equation turns into,

$$\|l\|' \|x\|' \leq \|lx\|.$$

Using the independence of choices of norms defining $|\cdot|_L$, we have,

$$|l|_L |x|_L \leq |lx|_L \leq |l|_L |x|_L.$$

Thus $|\cdot|_L$ is such norm.

Problem 6

We have $|1|_L = 1, |0|_L = 0$. It is also multiplicative. As it is a K -norm, we see this is ultrametric.

Problem 7

Since X is spectral $\mathfrak{Q}_c$ is a topology base of it. Note that for any $z \in S_x$, we have $x \in \Omega \in \mathfrak{Q}_C(X)$ then $z \in \Omega$. Thus,

$$S_x = \bigcap_{\substack{\Omega \in \mathfrak{Q}_C(X) \\ x \in \Omega}} \Omega.$$

From this

$$S_x \cap S_y = \bigcap_{\substack{\Omega \in \Omega_C(X) \\ x \in \Omega}} \Omega \cap \bigcap_{\substack{\Theta \in \Omega_C(X) \\ y \in \Theta}} \Theta = \bigcap_{\substack{x \in \Omega \\ y \in \Theta}} \Omega \cap \Theta.$$

Problem 8

Note that X_{con} is compact. By the assumption,

$$\bigcap_{\Omega, \Theta} \Omega \cap \Theta = \emptyset.$$

In X_{con} we get,

$$\bigcup (X \setminus (\Omega \cap \Theta)) = X_{\text{con}}$$

Using the compactness of X_{con} , for some $\Omega_i, \Theta_i, i = 1, \dots, n$,

$$\bigcap_{i=1}^n \Omega_i \cap \Theta_i = \emptyset.$$

In particular $\Omega = \bigcap_{i=1}^n \Omega_i, \Theta = \bigcap_{i=1}^n \Theta_i$ are disjoint sets separating x, y .

Problem 9

Since H is a subgroup $1 \in H$ and $0 \notin H$. Thus $w(1) = 1, w(0) = 0$. For the multiplicativeness, note that H is a group thus $x \in H, y \notin H \Rightarrow xy \notin H$ by the cancellation law. Thus w is multiplicative. w satisfies the strong triangle inequality on H and since it is 0 outside H , it holds globally on Γ .